28/10/26

Signal Processing

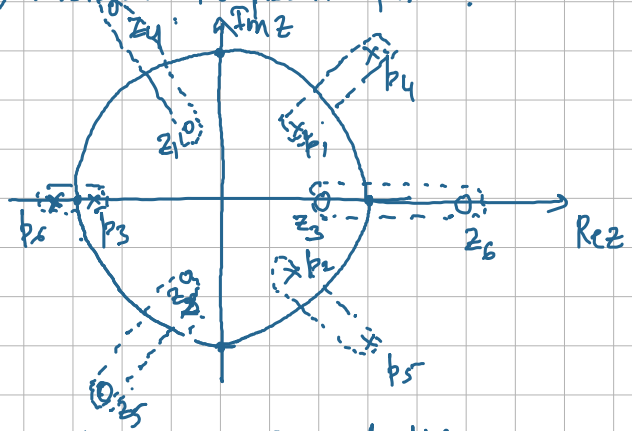
- Recap:- Inverse Systems, All-pass Systems, Min-phase Systems
  - Causal + stable systems (rational system functions)
- Relating magnitude and phase response ✓
  - | |, x, o; ∠, x, o, min-phase
  - $|H(e^{j\omega})|^2$
  - Examples

1: • $H_1(z) = \left[2(1-z^{-1})(1+0.5z^{-1})\right] / \left[(1-0.8e^{j\pi/4}z^{-1})(1-0.8e^{-j\pi/4}z^{-1})\right]$

• $H_2(z) = \left[(1-z^{-1})(1+2z^{-1})\right] / \left[(1-0.8e^{j\pi/4}z^{-1})(1-0.8e^{-j\pi/4}z^{-1})\right]$

Show that $C_1(z) = C_1(z)$.

2: C(z) with poles and zeros marked below. If system is stable, causal, and a_k, b_k are real, what are the possible options?



- Recall: $H(z) = \left(\frac{b_0}{a_0}\right) \frac{\prod_{k=1}^{M}(1-c_k z^{-1})}{\prod_{k=1}^{N}(1-d_k z^{-1})}$ (rational system)

  Assuming b_0, a_0 are real.

  $\prod_{k=1}^{M}(1-c_k z^{-1})$ ← zeros of H(z)

  $\prod_{k=1}^{N}(1-d_k z^{-1})$ ← poles of H(z)

- Recall: $H_{ap}(z) = A \cdot \prod_{k=1}^{N_r} \frac{(z^{-1}-d_k)}{(1-d_k z^{-1})} \prod_{k=1}^{N_c} \frac{(z^{-1}-e_k^*)}{(1-e_k z^{-1})} \frac{(z^{-1}-e_k)}{(1-e_k^* z^{-1})}$ — (*)

  N_r: real; N_c: complex

  $\frac{(z^{-1}-d_k)}{(1-d_k z^{-1})}$: Real

  $\frac{(z^{-1}-e_k^*)}{(1-e_k z^{-1})} \frac{(z^{-1}-e_k)}{(1-e_k^* z^{-1})}$: complex zero and poles

- We will show that the group delay $\text{grd}[H_{ap}(e^{j\omega})] \geq 0$ $\left\{ \text{grd}[H(e^{j\omega})] = -\frac{d}{d\omega} \angle \frac{H(e^{j\omega})}{ap} \right\}$
- We will show that the phase of $H_{ap}(e^{j\omega}) \leq 0$.
- As a consequence $\angle H(e^{j\omega}) = \angle H_{min}(e^{j\omega}) + \angle H_{ap}(e^{j\omega})$ — ①

$$\text{grd}[H(e^{j\omega})] = \text{grd}[H_{min}(e^{j\omega})] + \text{grd}[H_{ap}(e^{j\omega})] \quad — ②$$

- $H_{ap}(z) = \dfrac{z^{-1} - a^*}{1 - az^{-1}}$; • let $z = e^{j\omega}$, $a = re^{j\theta}$, stability, causality

Left side of class: find $\angle \frac{H(e^{j\omega})}{ap}$

Right side of class: find $\text{grd}[H_{ap}(e^{j\omega})]$

$$\angle H_{ap}(e^{j\omega}) = \tan^{-1} \frac{\text{Im}\{H_{ap}(e^{j\omega})\}}{\text{Re}\{H_{ap}(e^{j\omega})\}}$$

$$H_{ap}(e^{j\omega}) = \frac{e^{-j\omega} - re^{j\theta}}{1 - re^{j\theta}e^{-j\omega}} = \frac{e^{-j\omega}\left[1 - re^{j\omega}e^{-j\theta}\right]}{\left[1 - re^{j\theta}e^{-j\omega}\right]}$$

$$= e^{-j\omega}\frac{\left[1 - re^{-j(\theta-\omega)}\right]}{\left[1 - re^{j(\theta-\omega)}\right]} = \frac{e^{-j\omega}\left[1 - re^{-j(\theta-\omega)}\right]}{\left[1 - r\cos(\theta-\omega) - jr\sin(\theta-\omega)\right]}$$

$$= e^{-j\omega}\frac{\left[1 - r\cos(\theta-\omega) + jr\sin(\theta-\omega)\right]}{\left[1 - r\cos(\theta-\omega) - jr\sin(\theta-\omega)\right]}$$

$$= e^{-j\omega}\left[\frac{a}{a^*}\right] = e^{-j\omega}\frac{a \cdot a}{|a|^2}$$

$$= e^{-j\omega}\cdot\frac{\left[(1 - r\cos(\theta-\omega))^2 + j^2 r^2 \sin^2(\theta-\omega) + 2(1 - r\cos(\theta-\omega)) \cdot jr \cdot \sin(\theta-\omega)\right]}{|a|^2}$$

$$= e^{-j\omega}\left[1 + r^2\cos^2(\theta-\omega) - 2r\cos(\theta-\omega) - r^2\sin^2(\theta-\omega) + 2j\left(1 - r\cos(\theta-\omega)\right) r\sin(\theta-\omega)\right] / |a|^2$$

$$= \underbrace{e^{-j\omega}}\left[1 + r^2\left(\cos^2(\theta-\omega) - \sin^2(\theta-\omega)\right) - 2r\cos(\theta-\omega) + 2j\, r\cdot(1-\cos(\theta-\omega)\cdot\sin(\theta-\omega)\right] / |a|^2$$

$$\Rightarrow \angle H_{ap}(e^{j\omega}) = -\omega \overset{?}{\oplus} 2\tan^{-1}\left[\frac{r\sin(\theta-\omega)}{1 - r\cos(\theta-\omega)}\right] \overset{0}{=} \overset{0}{=}$$

Is $\text{grd}\left[H_{ap}(e^{j\omega})\right] = -\frac{d}{d\omega}\angle H_{ap}(e^{j\omega})$ $\boxed{> 0}$

$$-\frac{d}{d\omega}\left(-\omega + 2\tan^{-1}\left(\frac{r\sin(\theta-\omega)}{1-r\cos(\theta-\omega)}\right)\right) = +\infty\left[-1 - \frac{1}{1+(\quad)^2}\cdot\left(\frac{+r^2}{+r^2}\right)\right]$$

HW

- Importantly, $r < 1$.

  If indeed $\text{grd}\left[H_{ap}(e^{j\omega})\right] > 0$, and we know,

  $$\text{grd}\left[H(e^{j\omega})\right] = \text{grd}\left[H_{min}(e^{j\omega})\right] + \text{grd}\left[H_{ap}(e^{j\omega})\right] \quad (3)$$

  $\Rightarrow$ For all rational LTI systems with the same magnitude response, $H_{min}(e^{j\omega})$ has the min group delay.

- How do we show min phase delay?

  Claim: All pass systems have negative phase.

  Show this: Recall $\text{grd}\left[H_{ap}(e^{j\omega})\right] = -\frac{d}{d\omega}\angle H_{ap}(e^{j\omega})$ ✓

  $$\Rightarrow \angle H_{ap}(e^{j\omega}) = -\left[\int_0^{\omega} \underbrace{\text{grd}\left[H_{ap}(e^{j\omega})\right]}\, d\omega - \underbrace{\angle H_{ap}(0)}_{=0}\right] \overset{*}{-}$$

Since $\text{grd}[H(e^{j\omega})] > 0$, $\int_0^{\omega} \text{grd}[H_{ap}(e^{j\omega})]\, d\omega > 0$ ↓ due to the definition of $H_{ap}(z)$ in (*)

$\Rightarrow \angle H_{ap}(e^{j\omega}) < 0$

Now $\angle H(e^{j\omega}) = \angle H_{min}(e^{j\omega}) + \angle H_{ap}(e^{j\omega})$

$\Rightarrow$ of all rational LTI systems with the same magnitude response, $H_{min}(e^{j\omega})$ has minimum phase-lag

$H_{ap}(z) = \dfrac{(z^{-1} - \frac{1}{2})}{(1 - \frac{1}{2}z^{-1})}$ valid ap system with causality and stability

Find $\angle H_{ap}(e^{j0}) = 0$

- Shown why $H_{min}(z)$ is called min-phase.
- Shown why $H_{ap}(z)$ and $H_{min}(z)$ are important.
- Q: What can we say about the relation between the magnitude and phase of rational LTI systems?

A: In general, not much. However, when we add constraints like causality, stability, we can say something useful.